

1. Which of the following is FALSE:

A human gene can express at most 12 splice variants.

A codon is a nucleotide triplet that is translated into one amino acid.

The coding region with a protein-coding gene is used as the template for forming a protein.

Alternative splicing is a common phenomenon in both animals and plants.

2. Which of the following is FALSE about the organization of a eukaryotic gene:

The number of introns in a transcript is one less than the number of exons.

Genes that have only one exon are not alternatively spliced.

The length of an intron cannot be a multiple of 3.

Some eukaryotic gene transcripts can consist of a single exon.

3. What programs could you use to align RNA-seq reads to: i) a reference genome, and ii) a transcript database?

bowtie, cufflinks

tophat, bwa

bowtie, bcftools

bowtie, bwa

tophat, bowtie

tophat, split

tophat, cufflinks

4. Which of the following is FALSE:

As measures of gene expression, RPKM is determined at the level of reads and FPKM is determined at the level of fragments.

The sums of FPKMs of all transcripts of a gene is equal to the gene's expression level.

The sums of FPKM values of all genes in a sample is 1,000,000.

FPKM stands for fragments-per-kilobase of cDNA sequence-per million reads.

5. What programs could be used to: i) assemble transcripts from RNA-seq reads, and ii) identify potentially novel transcripts and genes

cufflinks, cuffcompare

tophat, samtools

tophat, cuffcompare

tophat, bowtie

6. Which of the following is FALSE about the gene annotations in the following GTF snippet:

```
chr1    MGF gene      3413609 3671498 .    -    .    gene_id "MG051951";
chr1    MGF transcript 3413609 3416344 .    -    .gene_id "MG051951"; transcrip
pt_id "MT162897";
chr1    MGF exon      3413609 3416344 .    -    .    gene_id "MG051951"; transcrip
t_id "MT162897";
chr1    MGF transcript 3421702 3671498 .    -    .    gene_id "MG051951"; trans
cript_id "MT070533";
chr1    MGF exon      3670552 3671498 .    -    .    gene_id "MG051951"; transcrip
t_id "MT070533";
chr1    MGF CDS 3670552 3671348 .    -    0    gene_id "MG051951"; transcript_id
"MT070533";
chr1    MGF exon      3421702 3421901 .    -    .    gene_id "MG051951"; transcrip
t_id "MT070533";
chr1    MGF CDS 3421792 3421901 .    -    1    gene_id "MG051951"; transcript_id
"MT070533";
```

The two transcripts for gene MG051951 overlap on the genome.

It contains only one gene, MG051951.

Gene MG051951 has two transcripts, MT162897 and MT070533.

Transcript MT162897 has a single exon.

Both exons of MT70533 contain both coding and non-coding sequences.

Transcript MT070533 is designated as coding.

Gene MG051951 is protein-coding.

Transcript MT070533 has 4 exons.

7. What does the following code NOT do:

```
BWT2IDX=/home/me/genomes/hg20/hg20
ANNOT=/home/me/genomes/hg20/myannot.gtf
ANNOTIDX=/home/me/genomes/hg20/myannot/myannot

mkdir -p /home/me/SRR100000
tophat2 -o /home/me/SRR100000 -p 10 --max-multihits 10 \
  -r 26 --mate-std-dev 25 \
  -a 6 \
  -G $ANNOT --transcriptome-index $ANNOTIDX \
  $BWT2IDX \
  /home/me/SRR100000_1.fastq.gz /home/me/SRR100000_2.fastq.gz
```

Load the gene annotations index from /home/me/genomes/hg20/myannot/myannot

Report spliced reads with at most 6 mismatches in the anchor site

Use information about the insert size distribution, e.g. its average (26) and standard deviation (25)

Use known gene annotations in the first mapping stage

Create the output in the /home/me/SRR100000 directory

Run multi-threaded, with 10 threads

Report only reads with 10 or fewer alignments on the genome

8. What does the following code NOT do:

```
TOPHATDIR=/home/florea/Tophat/
```

```
mkdir -p Test1
```

```
cd Test1
```

```
ln -s $TOPHATDIR/accepted_hits.bam .
```

```
cufflinks -L Test1 -p 8 -j 0.10 -F 0.05 accepted_hits.bam
```

Report all isoforms whose abundance is 0.05 or more of the most abundant isoform's for that gene

Use the default reference transcript annotation to guide assembly

Generate a file named transcripts.gtf in directory Test1, containing all assembled transfrags

Use 0.10 as the minimum level for intra-intronic transcripts

9. Which of the following is NOT described in the following summary file produced by tophat:

Left reads:

Input : 60586968
Mapped : 58163843 (96.0% of input)
of these: 6832240 (11.7%) have multiple alignments (359075 have
>10)

Right reads:

Input : 60586968
Mapped : 56969290 (94.0% of input)
of these: 6668479 (11.7%) have multiple alignments (358573 have
>10)

95.0% overall read mapping rate.

Aligned pairs: 55880048

of these: 6491876 (11.6%) have multiple alignments
2795712 (5.0%) are discordant alignments

87.6% concordant pair alignment rate.

Of the mapped mate 2 reads, 5.0% had multiple matches on the genome

The library was strand-specific

94.0% of the mate 2 reads were mapped

Of the mapped mate 1 reads, 11.7% had multiple matches on the genome

The reads were 100 bp long

The number of read pairs in which both mates were aligned is 55,880,048

Of the aligned pairs, 6,491,876 had multiple matches on the genome

Of the pairs in which both reads were aligned, 5.0% were at a distance or in an orientation inconsistent with the insert size and/or library-type

10. Which of the following is NOT TRUE about the output below, obtained from a cuffdiff differential expression analysis:

```
XLOC_000002 XLOC_000002 AT1G01020 1:5927-  
8737 q1 q2 OK 1.13032 3.48406 1.62404 0.694576 0.5277 0.998846 no  
XLOC_000004 XLOC_000004 AT1G01073 1:44676-  
44787 q1 q2 NOTEST 0 0 0 0 1 1 no  
XLOC_000042 XLOC_000042 AT1G01580 1:209394-213041 q1 q2 OK 1.59512 0 -  
inf nan 5e-05 0.0096703 yes
```

There are not enough alignments for testing for differential expression at locus
XLOC_000004

**There are too many alignments for testing for differential expression at locus
XLOC_000004**

Locus XLOC_000004 corresponds to gene AT1G01073

Locus XLOC_000042 corresponds to gene AT1G01580